

Experiment 3

Design of Test Cases and a Maintenance Plan for an AI Study Assistant Application

Aim

To design test cases and prepare a maintenance plan for the AI Study Assistant application to ensure software quality, reliability, and continuous improvement.

Objective

- Design test cases for key application features.
- Verify the expected behaviour of the system.
- Prepare a software maintenance plan.
- Ensure application reliability and performance.

1. Introduction

Testing is an important phase in SDLC that verifies whether the application works according to the specified requirements. After deployment, maintenance is carried out to fix bugs, improve performance, and add new features based on user feedback.

2. Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	User Registration	Account created successfully	Pass
TC-02	User Login	Login successful with valid credentials	Pass
TC-03	Upload PDF Notes	File uploaded successfully	Pass
TC-04	Ask AI Question	AI displays a relevant answer	Pass
TC-05	Generate Quiz	Quiz generated based on uploaded notes	Pass
TC-06	View Progress Dashboard	Progress details displayed correctly	Pass
TC-07	Invalid Login	Error message displayed	Pass
TC-08	Logout	User logged out successfully	Pass

3. Types of Testing

- 1) Unit Testing
- 2) Integration Testing
- 3) System Testing
- 4) User Acceptance Testing (UAT)

4. Defect Reporting

Defect	Priority	Solution
AI gives incorrect response	High	Improve prompt and AI model
PDF upload failure	Medium	Validate file format and size
Slow dashboard loading	Medium	Optimize database queries
Login issue	High	Fix authentication logic

5. Maintenance Plan

- ☐ Corrective Maintenance
 - ☐ Fix software bugs.
 - ☐ Resolve login and AI response issues.
- ☐ Adaptive Maintenance
 - ☐ Update the application for new technologies.
 - ☐ Support new browser and device versions.
- ☐ Perfective Maintenance
 - ☐ Improve AI response quality.
 - ☐ Enhance user interface.
 - ☐ Add new study features.
- ☐ Preventive Maintenance
 - ☐ Regular database backup.
 - ☐ Security updates.
 - ☐ Performance monitoring.

6. Maintenance Schedule

Activity	Frequency
Bug Fixes	As Required
Security Updates	Monthly
Database Backup	Weekly
Performance Check	Monthly
Feature Updates	Every 3 Months

7. Expected Outcome

The testing process ensures that the AI Study Assistant performs accurately and reliably. The maintenance plan helps keep the application secure, updated, and responsive to user needs.

8. Conclusion

Proper testing helps identify and fix software issues before deployment, while a structured maintenance plan ensures the application remains secure, reliable, and efficient. Together, they improve software quality and enhance the overall learning experience for students.